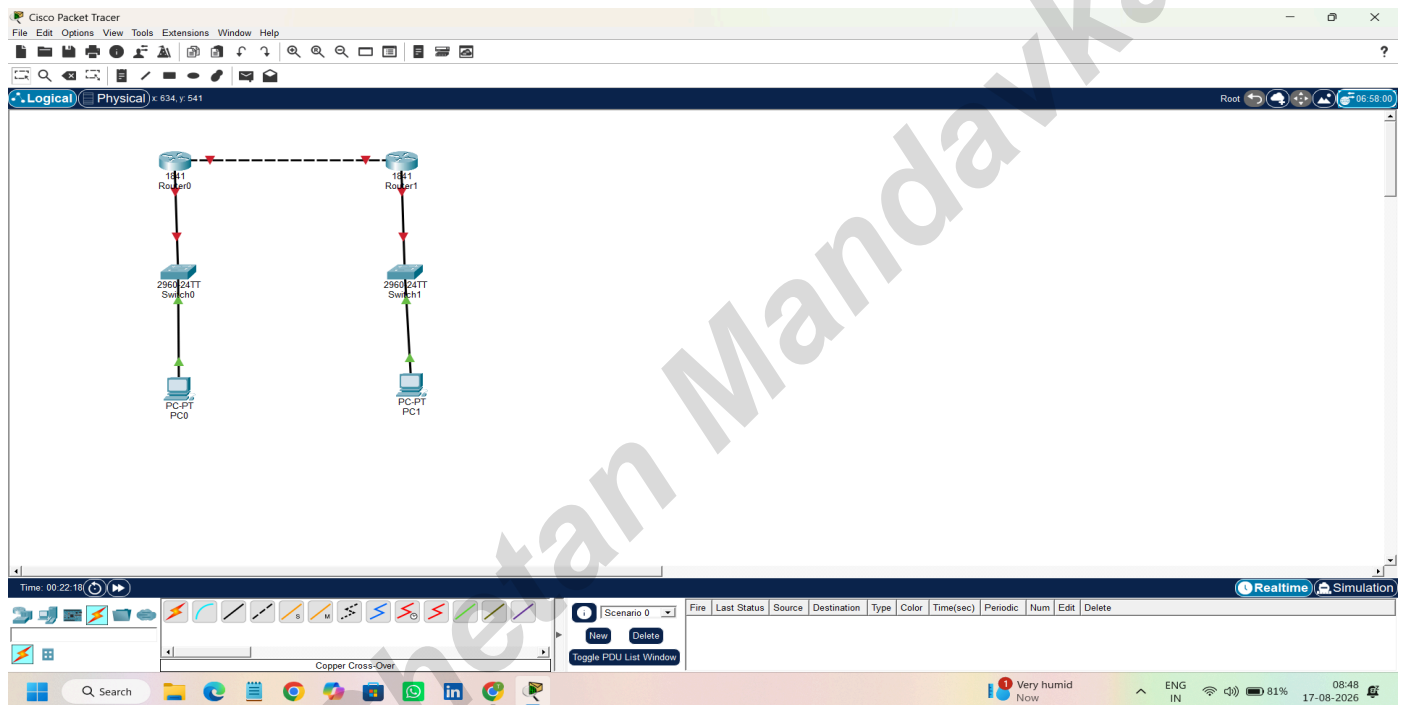


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Aim:- Configure IPsec on network devices to provide secure communication and protect against unauthorized access and attacks.

Step 1:- Open Cisco Packet And Add 2 Routers, 2 Switches, 2 PCs And Connect Them To Copper Straight Through And Router Connect With Copper Cross Over



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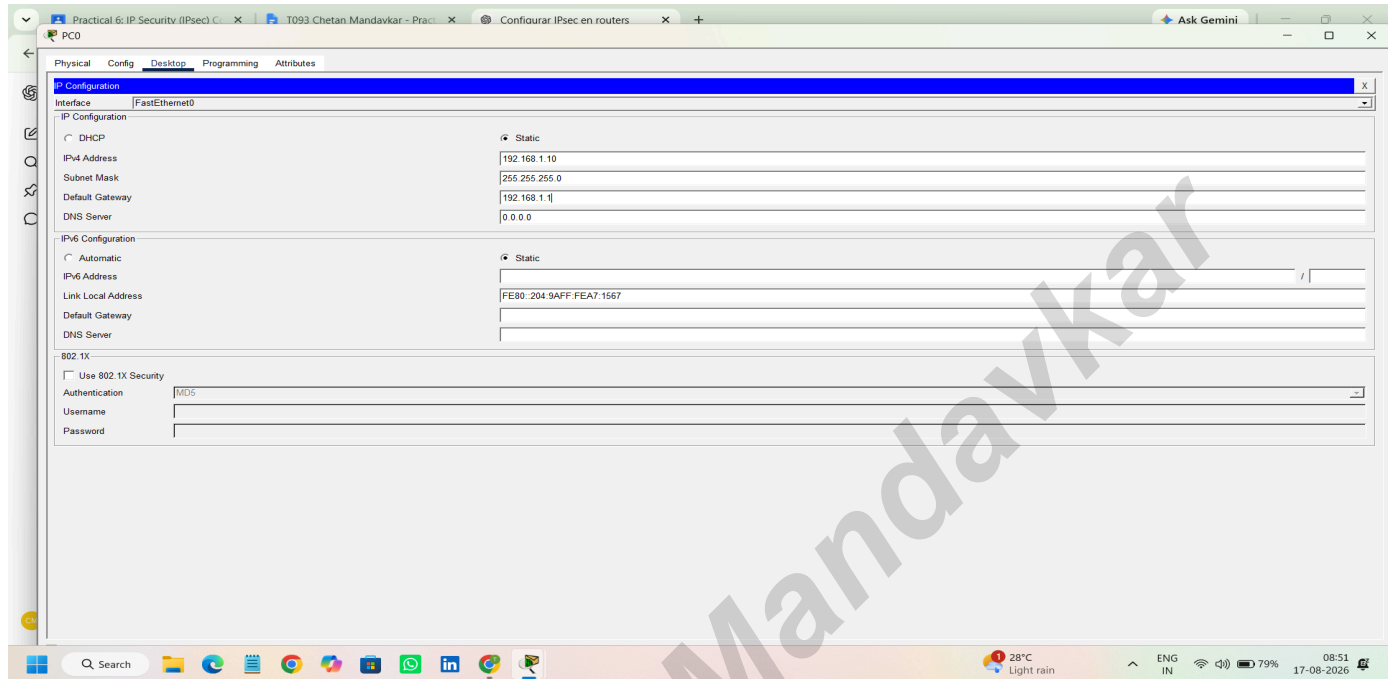
Roll no. T093

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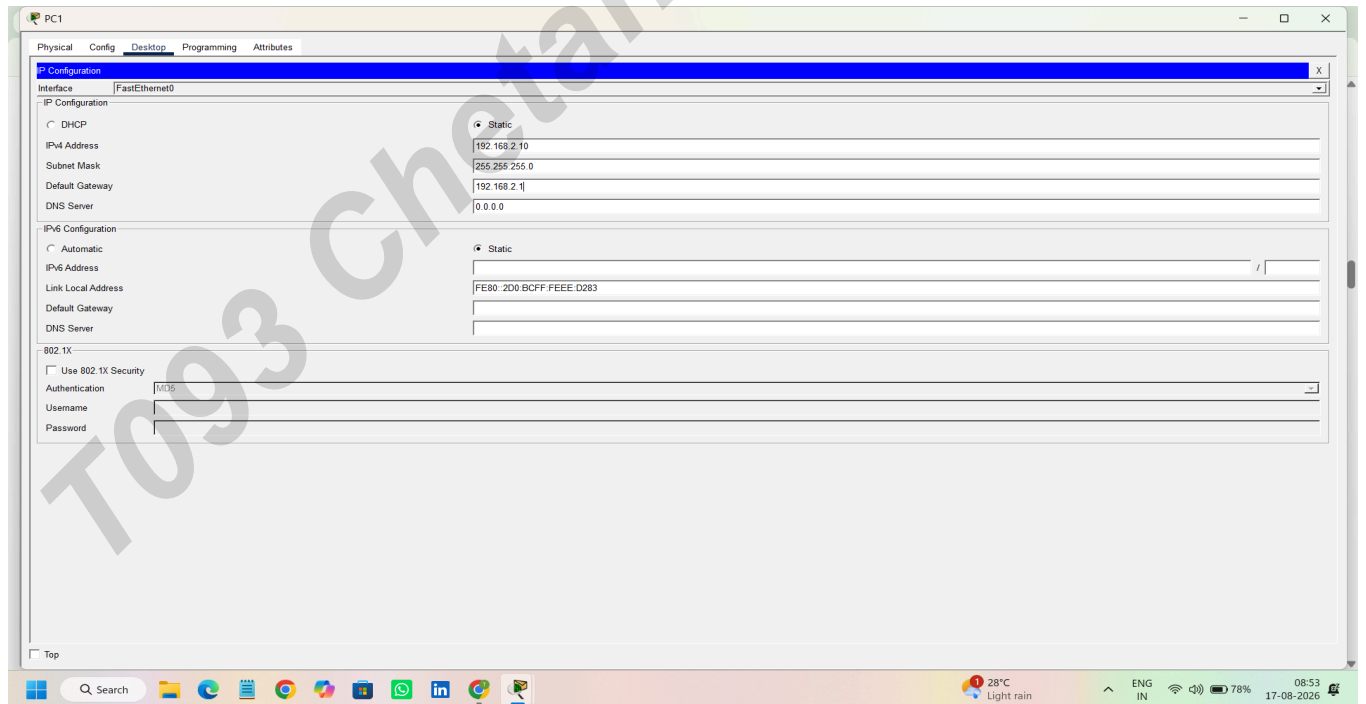
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Step 2:- Configure PCs

PC0:-



PC1:-



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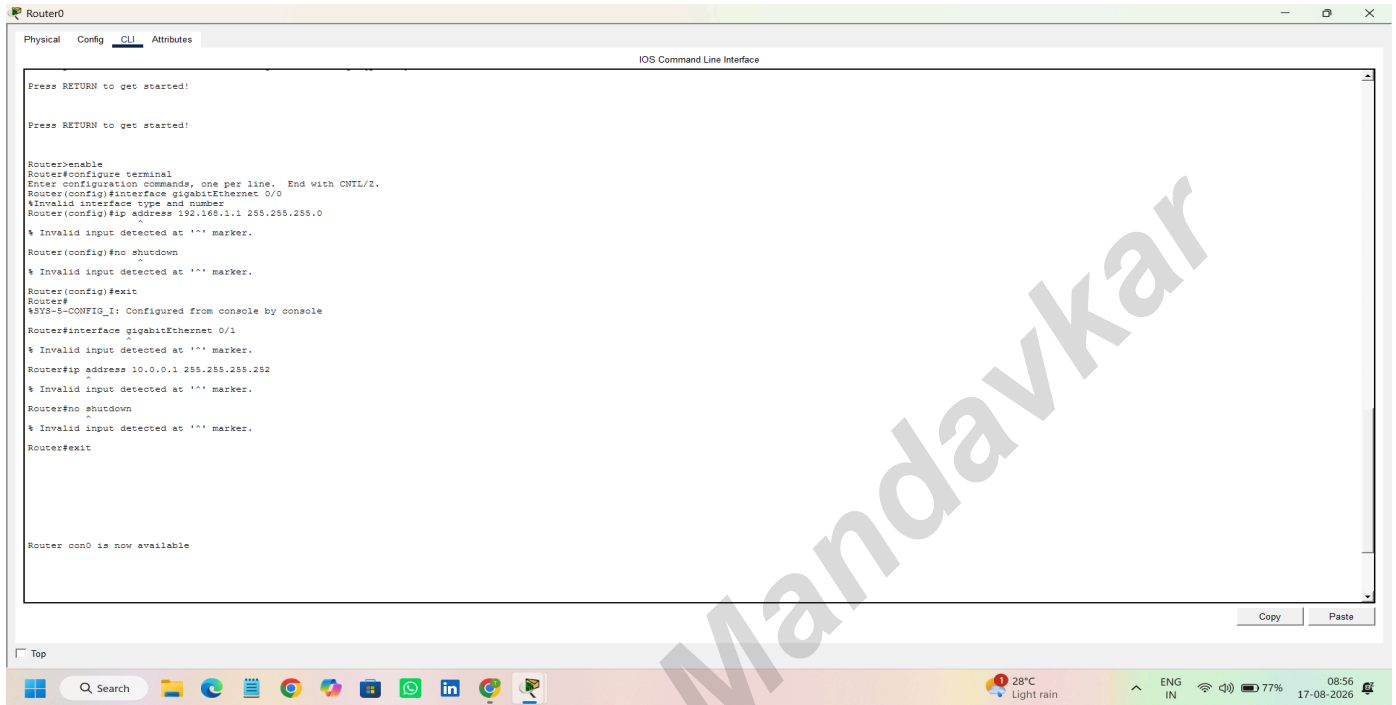
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Step 3:- Configure Routers

Router 0:-

A screenshot of a web-based network simulator interface for Router0. The window has tabs for Physical, Config, CLI, and Attributes, with CLI selected. The CLI window shows the IOS Command Line Interface. The user has entered commands to enable the router, enter configuration mode, configure interface gigabitEthernet 0/0 with IP address 192.168.1.1 and mask 255.255.255.0, and configure interface gigabitEthernet 0/1 with IP address 10.0.0.1 and mask 255.255.255.252. The router is now available. The background of the CLI window is watermarked with 'Chetan Mandavkar'.

```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

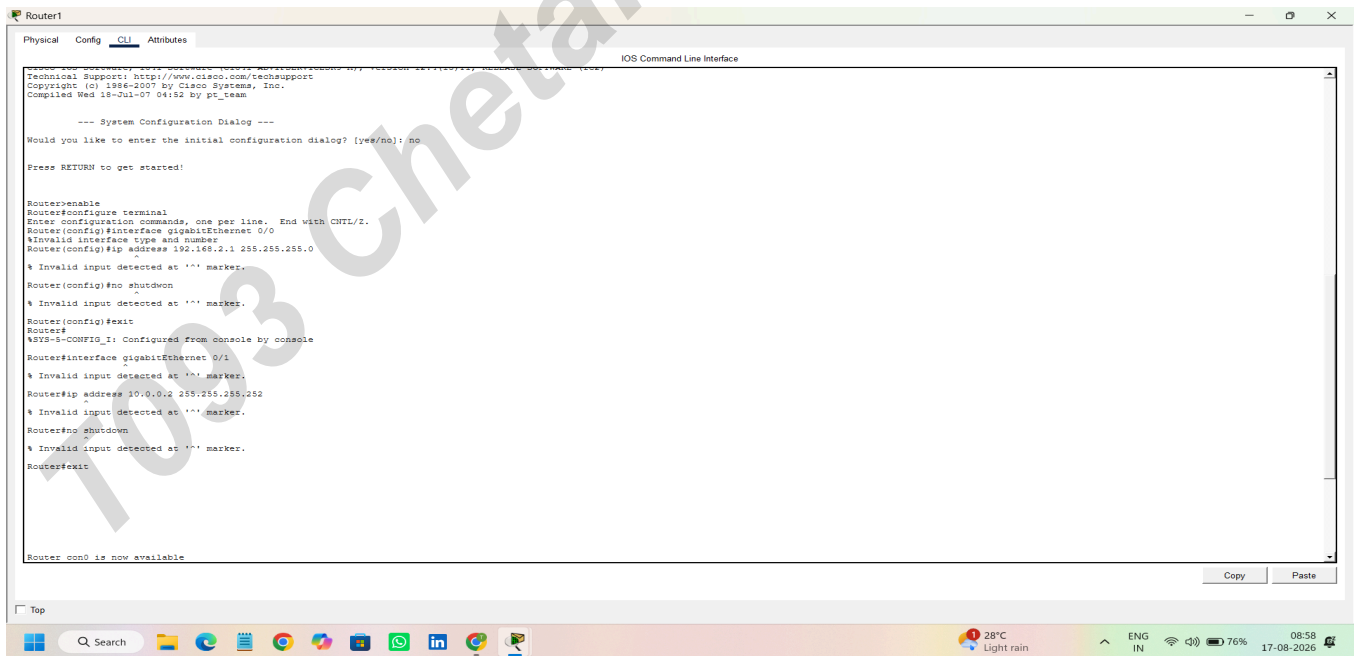
Press RETURN to get started!

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config)#ip address 192.168.1.1 255.255.255.0
% Invalid input detected at '^' marker.
Router(config)#no shutdown
% Invalid input detected at '^' marker.
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#interface gigabitEthernet 0/1
% Invalid input detected at '^' marker.
Router#ip address 10.0.0.1 255.255.255.252
% Invalid input detected at '^' marker.
Router#no shutdown
% Invalid input detected at '^' marker.
Router#exit

Router con0 is now available
```

Router 1:-

A screenshot of a web-based network simulator interface for Router1. The window has tabs for Physical, Config, CLI, and Attributes, with CLI selected. The CLI window shows the IOS Command Line Interface. The user has entered commands to enable the router, enter configuration mode, configure interface gigabitEthernet 0/0 with IP address 192.168.2.1 and mask 255.255.255.0, and configure interface gigabitEthernet 0/1 with IP address 10.0.0.2 and mask 255.255.255.252. The router is now available. The background of the CLI window is watermarked with 'Chetan Mandavkar'.

```
Router1
Physical Config CLI Attributes
IOS Command Line Interface

--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config)#ip address 192.168.2.1 255.255.255.0
% Invalid input detected at '^' marker.
Router(config)#no shutdown
% Invalid input detected at '^' marker.
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#interface gigabitEthernet 0/1
% Invalid input detected at '^' marker.
Router#ip address 10.0.0.2 255.255.255.252
% Invalid input detected at '^' marker.
Router#no shutdown
% Invalid input detected at '^' marker.
Router#exit

Router con0 is now available
```

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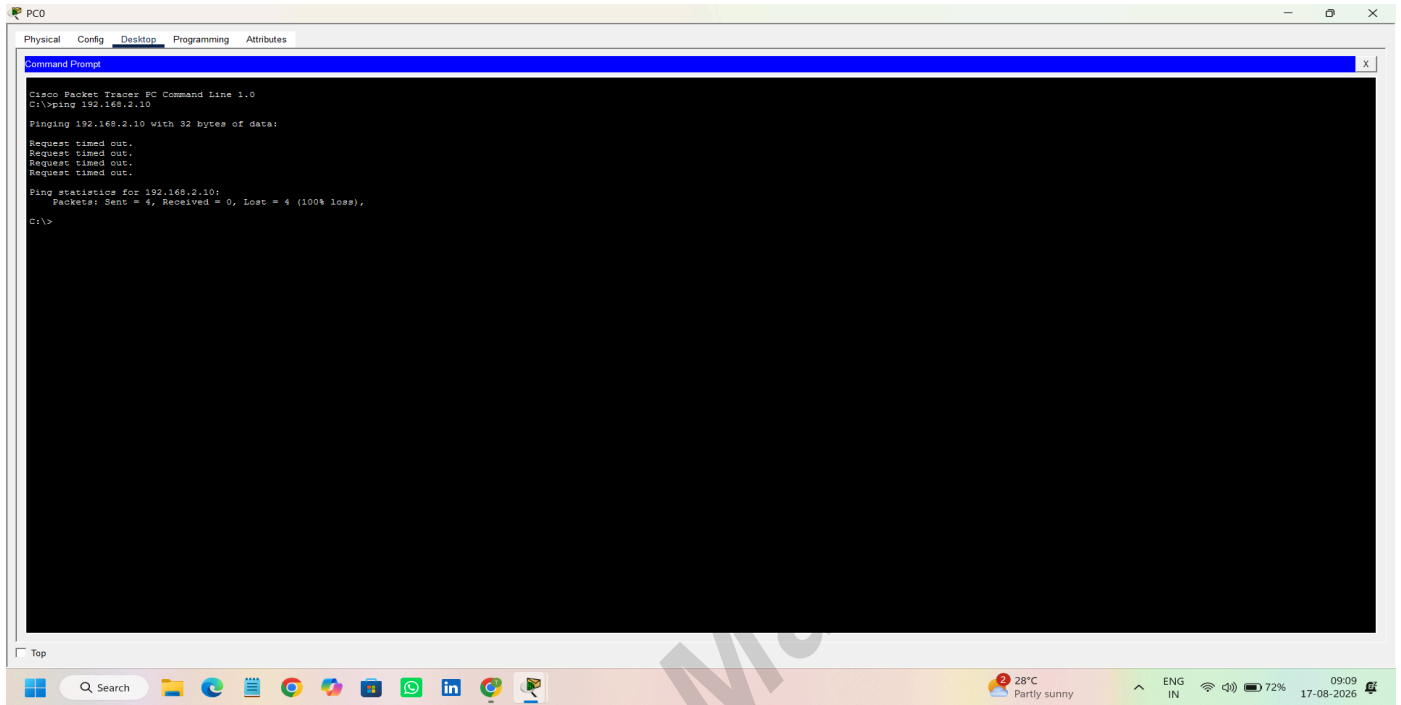
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Step 4:- Test basic connectivity

PC0:-



The screenshot shows a Windows 10 desktop environment. The 'PC0' window is open, displaying the 'Command Prompt' application. The command prompt shows the following output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.10

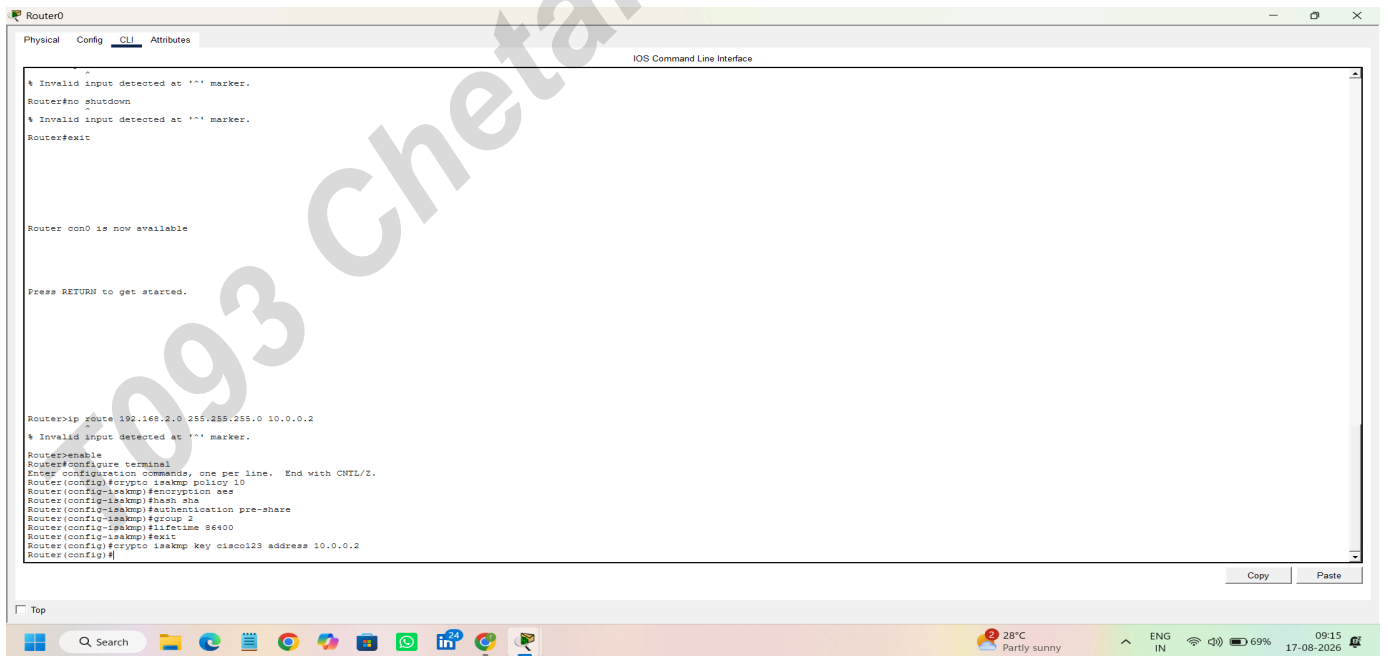
Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
```

The taskbar at the bottom shows the Windows Start button, a search bar, and several application icons. The system tray on the right indicates a temperature of 28°C, 'Partly sunny' weather, language set to 'ENG IN', and a battery level of 72% at 09:09 on 17-08-2026.

Step 5:- Configure IKE Phase 1 on Router0



The screenshot shows the 'Router0' window in Cisco Packet Tracer, displaying the 'IOS Command Line Interface'. The configuration commands entered are as follows:

```
Router>no shutdown
Router>exit

Router con0 is now available

Press RETURN to get started.

Router>ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router>enable
Router(config)#configure terminal
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#hash sha
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 2
Router(config-isakmp)#lifetime 86400
Router(config-isakmp)#exit
Router(config)#crypto isakmp key cisco123 address 10.0.0.2
Router(config)#
```

The taskbar at the bottom is identical to the previous screenshot, showing the same system information and application icons.

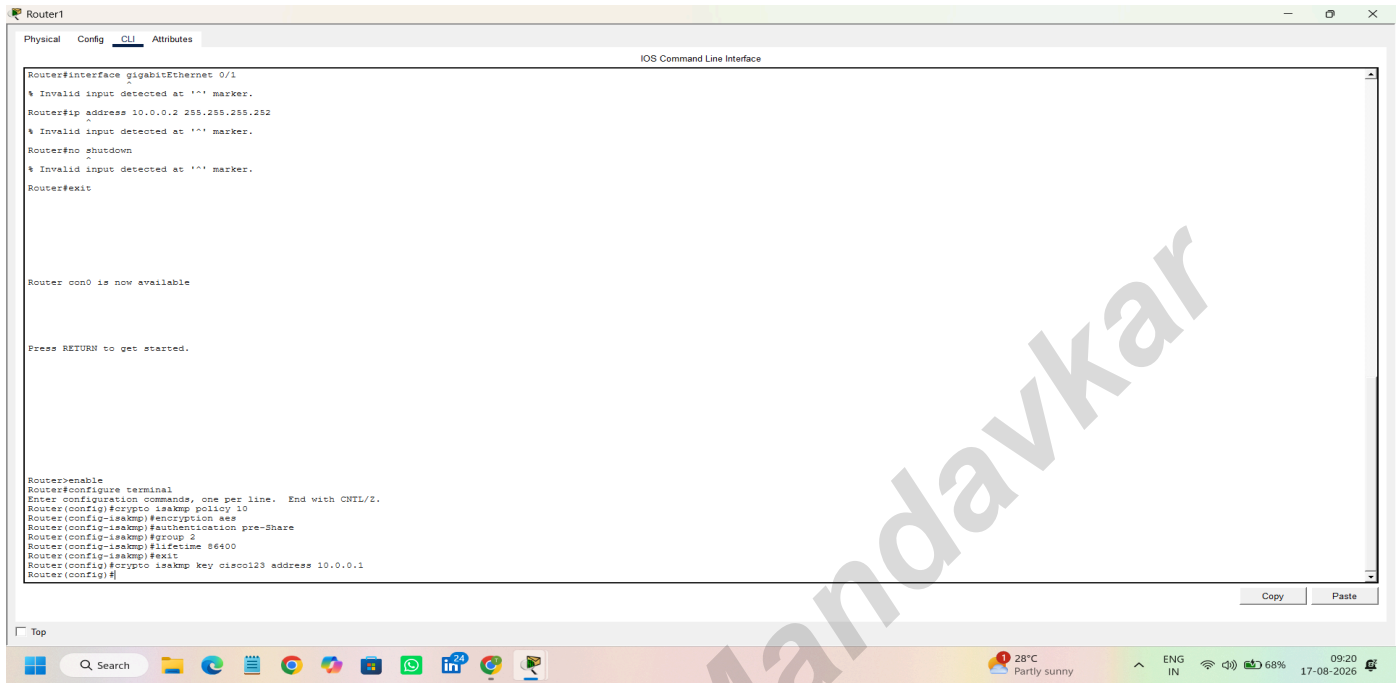
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Step 6:- Configure IKE Phase 1 on Router1



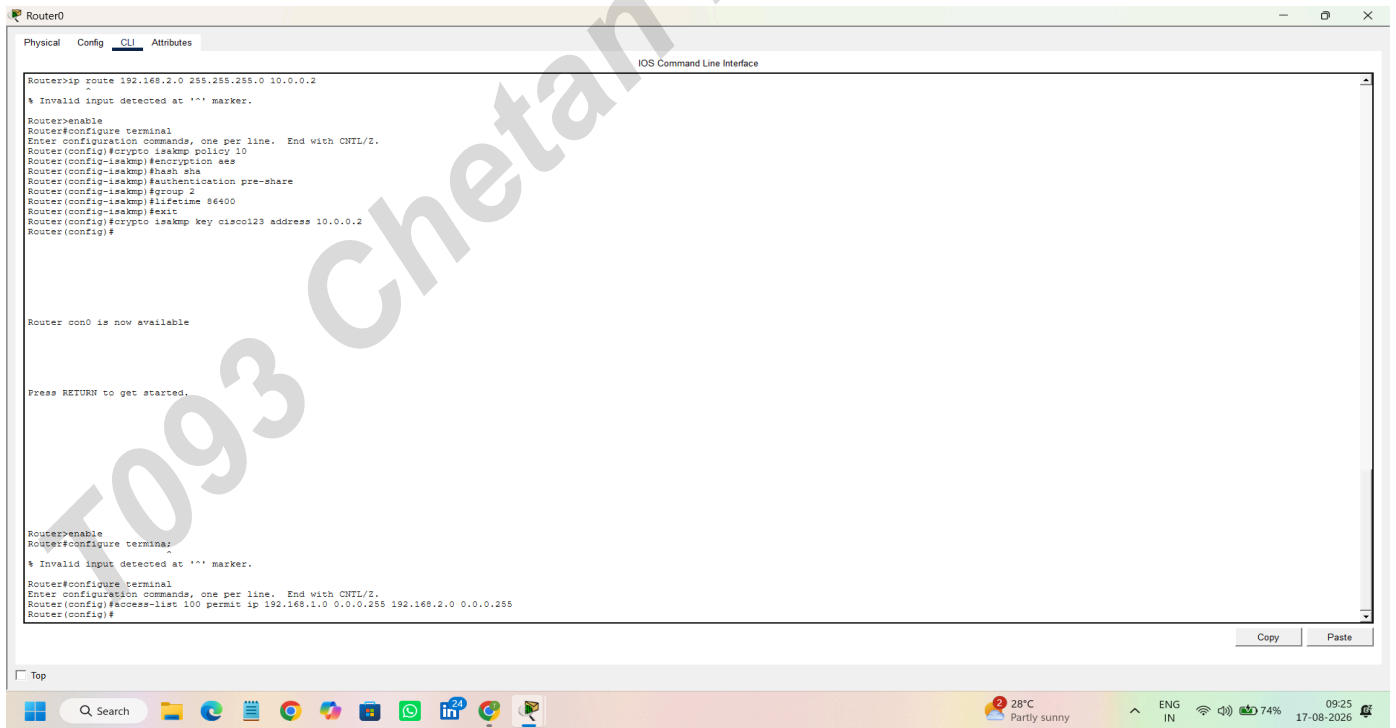
The screenshot shows the Cisco IOS Command Line Interface for Router1. The configuration includes setting the interface to gigabitEthernet 0/1, assigning the IP address 10.0.0.2 with a subnet mask of 255.255.255.252, and enabling the interface. The user then enters the 'enable' mode and configures the terminal. The configuration commands for IKE Phase 1 are as follows:

```
Router1#interface gigabitEthernet 0/1
Router1#ip address 10.0.0.2 255.255.255.252
Router1#no shutdown
Router1#exit

Router1#enable
Router1#configure terminal
Router1(config)#crypto isakmp policy 10
Router1(config-isakmp)#encryption aes
Router1(config-isakmp)#authentication pre-share
Router1(config-isakmp)#group 2
Router1(config-isakmp)#lifetime 86400
Router1(config-isakmp)#exit
Router1(config)#crypto isakmp key cisco123 address 10.0.0.1
Router1(config)#
```

The bottom of the window shows a Windows taskbar with various application icons, a search bar, and system status information including temperature (28°C), weather (Partly sunny), language (ENG IN), and battery level (68%).

Step 7:- Create the IPsec ACL on Router0



The screenshot shows the Cisco IOS Command Line Interface for Router0. The configuration includes setting the interface to gigabitEthernet 0/0, assigning the IP address 192.168.2.0 with a subnet mask of 255.255.255.0, and enabling the interface. The user then enters the 'enable' mode and configures the terminal. The configuration commands for IPsec ACL are as follows:

```
Router0#ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router0#enable
Router0#configure terminal
Router0(config)#crypto isakmp policy 10
Router0(config-isakmp)#encryption aes
Router0(config-isakmp)#hash sha
Router0(config-isakmp)#authentication pre-share
Router0(config-isakmp)#group 2
Router0(config-isakmp)#lifetime 86400
Router0(config-isakmp)#exit
Router0(config)#crypto isakmp key cisco123 address 10.0.0.2
Router0(config)#

Router0#enable
Router0#configure terminal
Router0(config)#access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
Router0(config)#
```

The bottom of the window shows a Windows taskbar with various application icons, a search bar, and system status information including temperature (28°C), weather (Partly sunny), language (ENG IN), and battery level (74%).

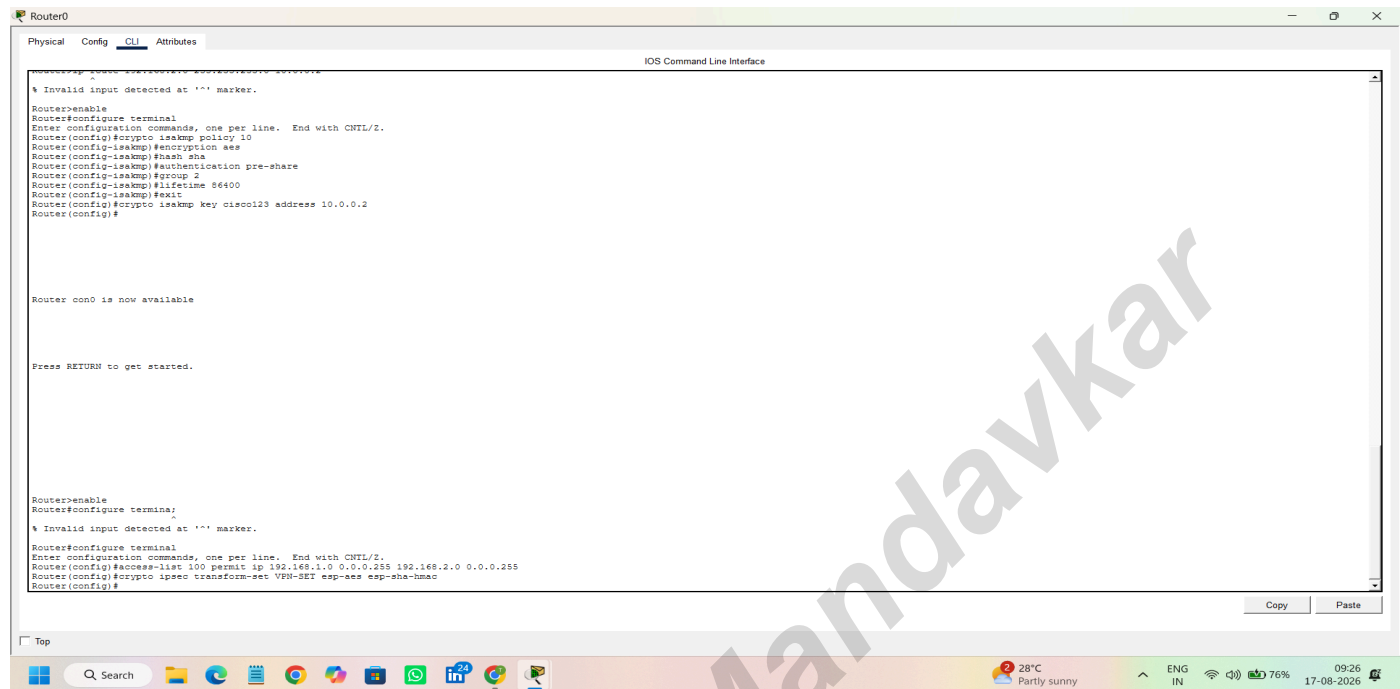
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Step 8:- Create the IPsec transform set on Router0

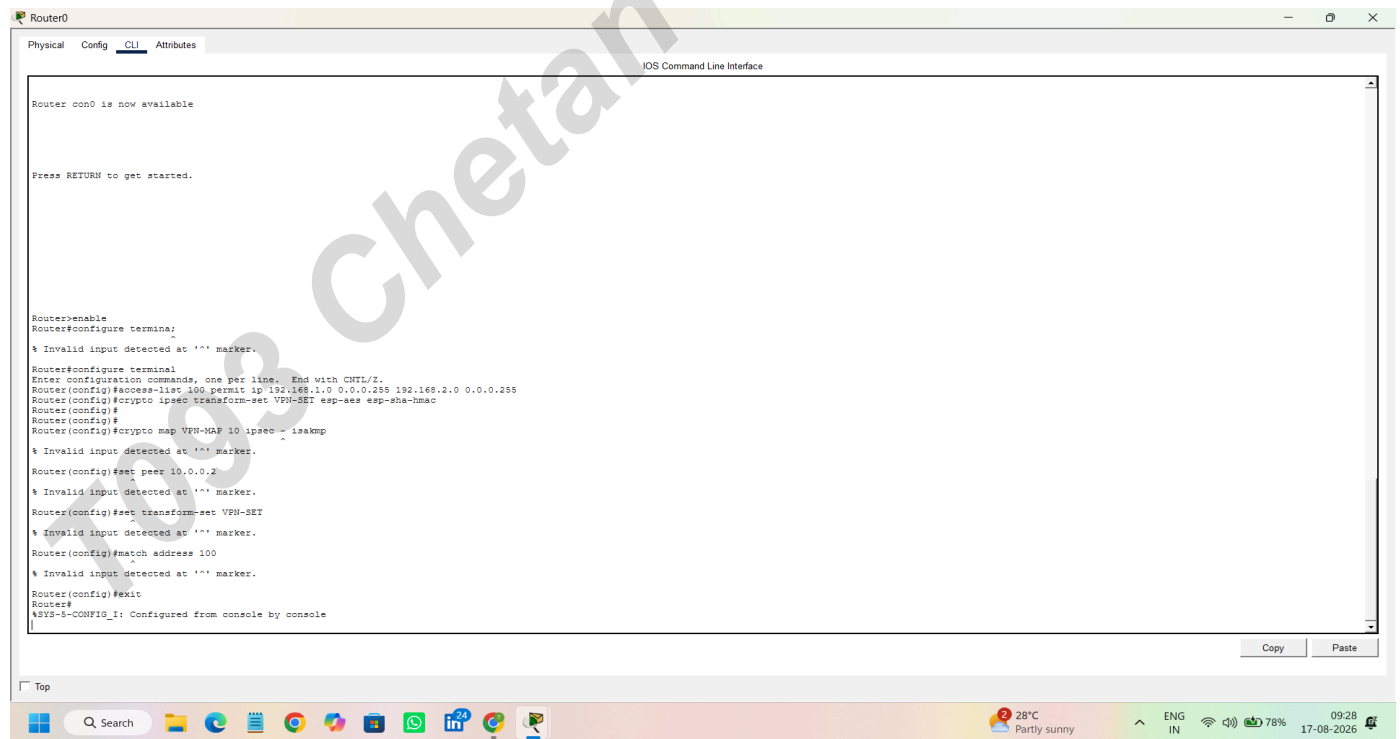


The screenshot shows the Cisco IOS Command Line Interface for Router0. The configuration steps for the IPsec transform set are as follows:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#hash sha
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 2
Router(config-isakmp)#lifetime 86400
Router(config-isakmp)#exit
Router(config)#crypto isakmp key cisco123 address 10.0.0.2
Router(config)#
```

Below the configuration, the status message "Router con0 is now available" is displayed. The bottom of the window shows a Windows taskbar with various application icons and system status information including temperature (28°C), network status, and battery level (76%).

Step 9:- Create the Crypto Map on Router0



The screenshot shows the Cisco IOS Command Line Interface for Router0. The configuration steps for the Crypto Map are as follows:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
Router(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Router(config)#
Router(config)#crypto map VPN-MAP 10 ipsec-isakmp
Router(config)#
Router(config)#set peer 10.0.0.2
Router(config)#set transform-set VPN-SET
Router(config)#match address 100
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

The configuration is complete, and the system message "%SYS-5-CONFIG_I: Configured from console by console" is shown. The bottom of the window shows a Windows taskbar with various application icons and system status information including temperature (28°C), network status, and battery level (78%).

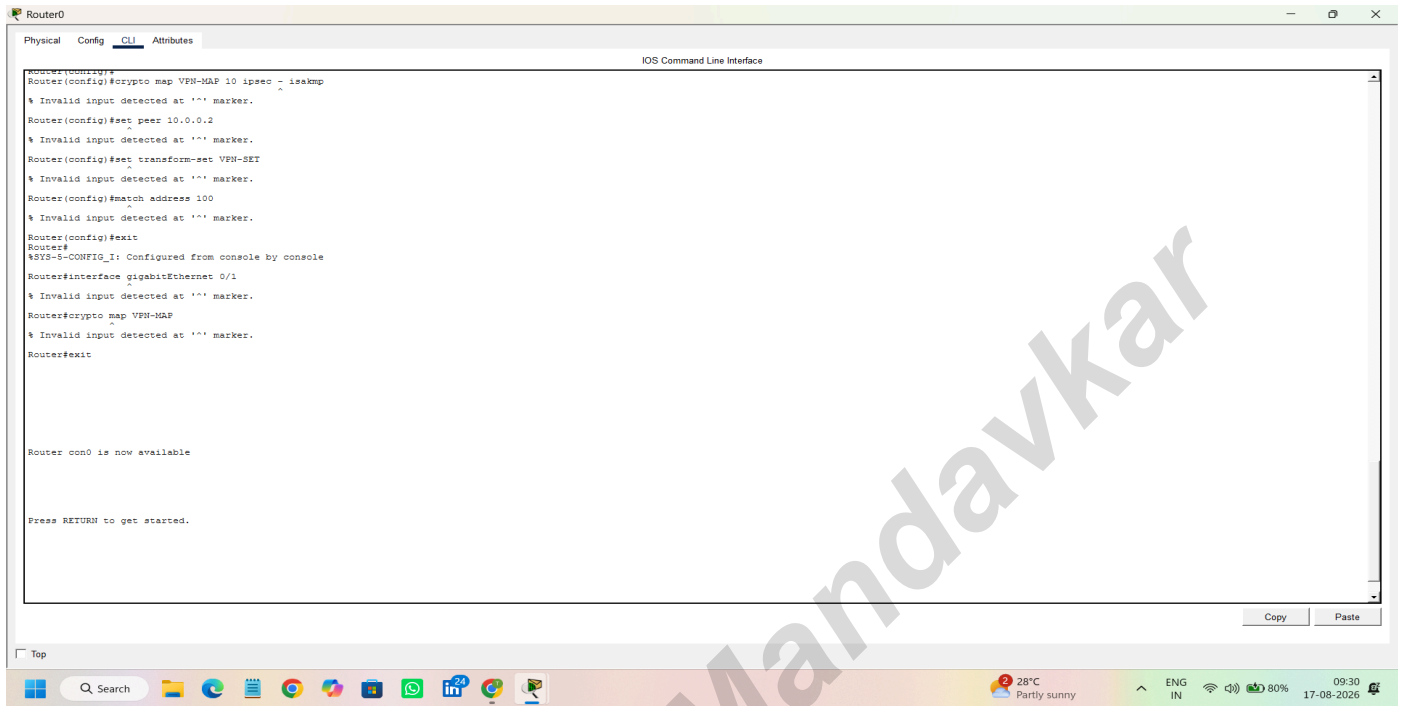
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Step 10:- Apply Crypto Map to Router0 WAN interface

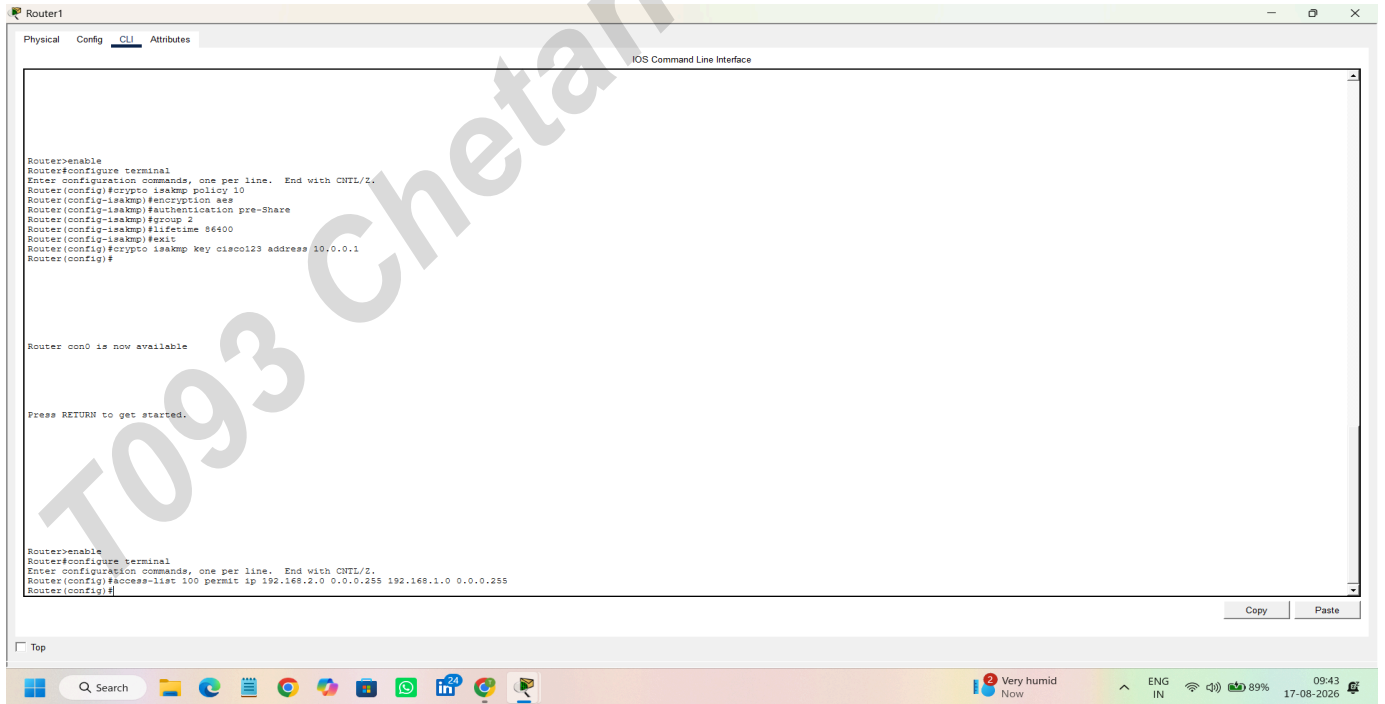
A screenshot of a Cisco IOS Command Line Interface (CLI) window for Router0. The window has tabs for Physical, Config, CLI, and Attributes, with CLI selected. The CLI shows a configuration session where a crypto map named 'VPN-MAP' is being applied to the GigabitEthernet 0/1 interface. The configuration includes setting the peer to 10.0.0.2, the transform-set to VPN-SET, and matching address 100. The session ends with 'Router#exit' and a message 'Router con0 is now available'. The Windows taskbar at the bottom shows various application icons, a search bar, and system status information including temperature (28°C), weather (Partly sunny), language (ENG IN), and time (09:30, 17-08-2026).

```
Router0>enable
Router0#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router0(config)#crypto map VPN-MAP 10 ipsec - isakmp
% Invalid input detected at '^' marker.
Router0(config)#set peer 10.0.0.2
% Invalid input detected at '^' marker.
Router0(config)#set transform-set VPN-SET
% Invalid input detected at '^' marker.
Router0(config)#match address 100
% Invalid input detected at '^' marker.
Router0(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#interface gigabitEthernet 0/1
% Invalid input detected at '^' marker.
Router#crypto map VPN-MAP
% Invalid input detected at '^' marker.
Router#exit

Router con0 is now available

Press RETURN to get started.
```

Step 11:- Configure IPsec ACL on Router1

A screenshot of a Cisco IOS Command Line Interface (CLI) window for Router1. The window has tabs for Physical, Config, CLI, and Attributes, with CLI selected. The CLI shows a configuration session where an IPsec ACL is being configured. The configuration includes enabling the terminal, setting the crypto isakmp policy to 10, setting the description to 'aaa', setting authentication to 'pre-Share', setting the group to 2, setting the lifetime to 86400, and setting the key to 'cisco123' for address 10.0.0.1. The session ends with 'Router#exit'. The Windows taskbar at the bottom shows various application icons, a search bar, and system status information including weather (Very humid Now), language (ENG IN), and time (09:43, 17-08-2026).

```
Router1>enable
Router1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router1(config)#crypto isakmp policy 10
Router1(config-isakmp)#description aaa
Router1(config-isakmp)#authentication pre-Share
Router1(config-isakmp)#group 2
Router1(config-isakmp)#lifetime 86400
Router1(config-isakmp)#exit
Router1(config)#crypto isakmp key cisco123 address 10.0.0.1
Router1(config)#

Router con0 is now available

Press RETURN to get started.

Router1>enable
Router1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router1(config)#access-list 100 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
Router1(config)#
```

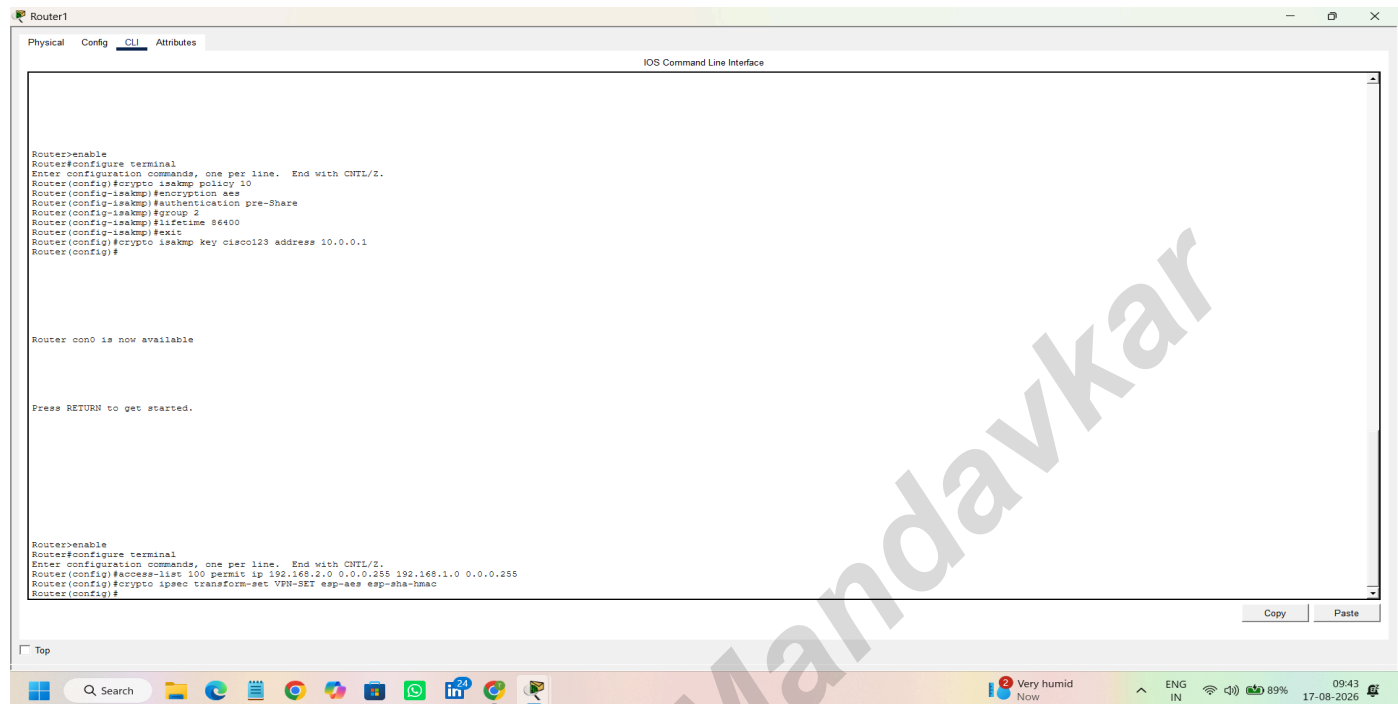
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Step 12:- Configure the Transform Set on Router1



```
Router1
Physical Config CLI Attributes
IOS Command Line Interface

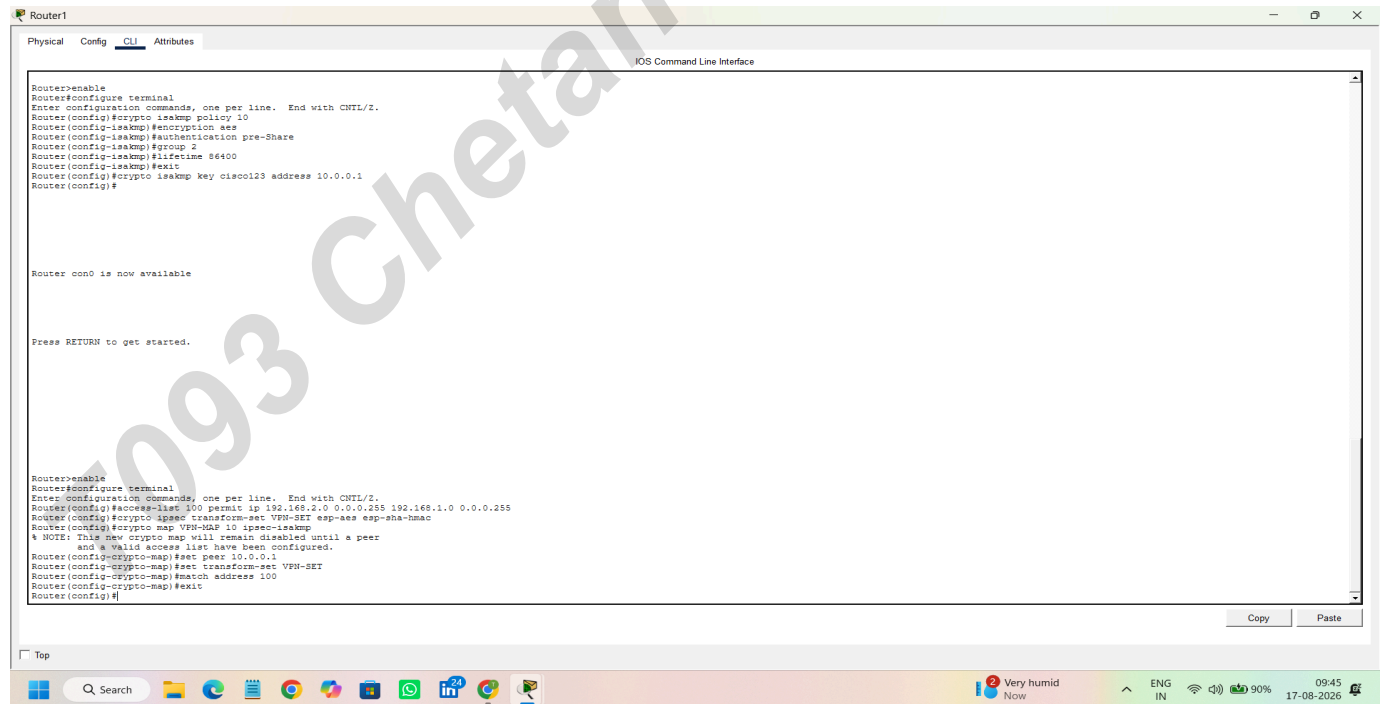
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#authentication pre-Share
Router(config-isakmp)#group 2
Router(config-isakmp)#lifetime 86400
Router(config-isakmp)#exit
Router(config)#crypto isakmp key cisco123 address 10.0.0.1
Router(config)#

Router con0 is now available.

Press RETURN to get started.

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 100 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
Router(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Router(config)#
```

Step 13:- Configure Crypto Map on Router1



```
Router1
Physical Config CLI Attributes
IOS Command Line Interface

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#authentication pre-Share
Router(config-isakmp)#group 2
Router(config-isakmp)#lifetime 86400
Router(config-isakmp)#exit
Router(config)#crypto isakmp key cisco123 address 10.0.0.1
Router(config)#

Router con0 is now available.

Press RETURN to get started.

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 100 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
Router(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Router(config)#crypto map VPN-MAP 10 ipsec-isakmp
! NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Router(config-crypto-map)#set peer 10.0.0.1
Router(config-crypto-map)#set transform-set VPN-SET
Router(config-crypto-map)#match address 100
Router(config-crypto-map)#exit
Router(config)#
```

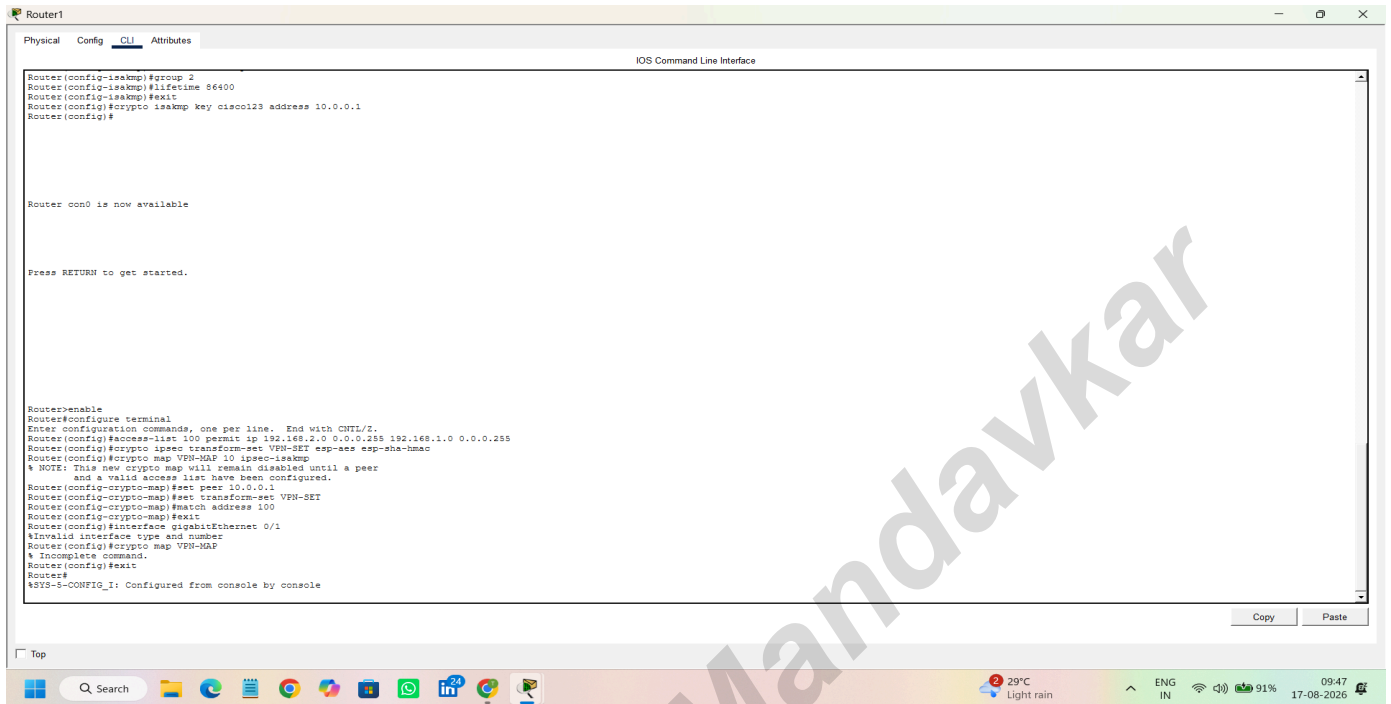
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Step 14:- Apply Crypto Map to Router1



```
Router1
Physical Config CLI Attributes
IOS Command Line Interface

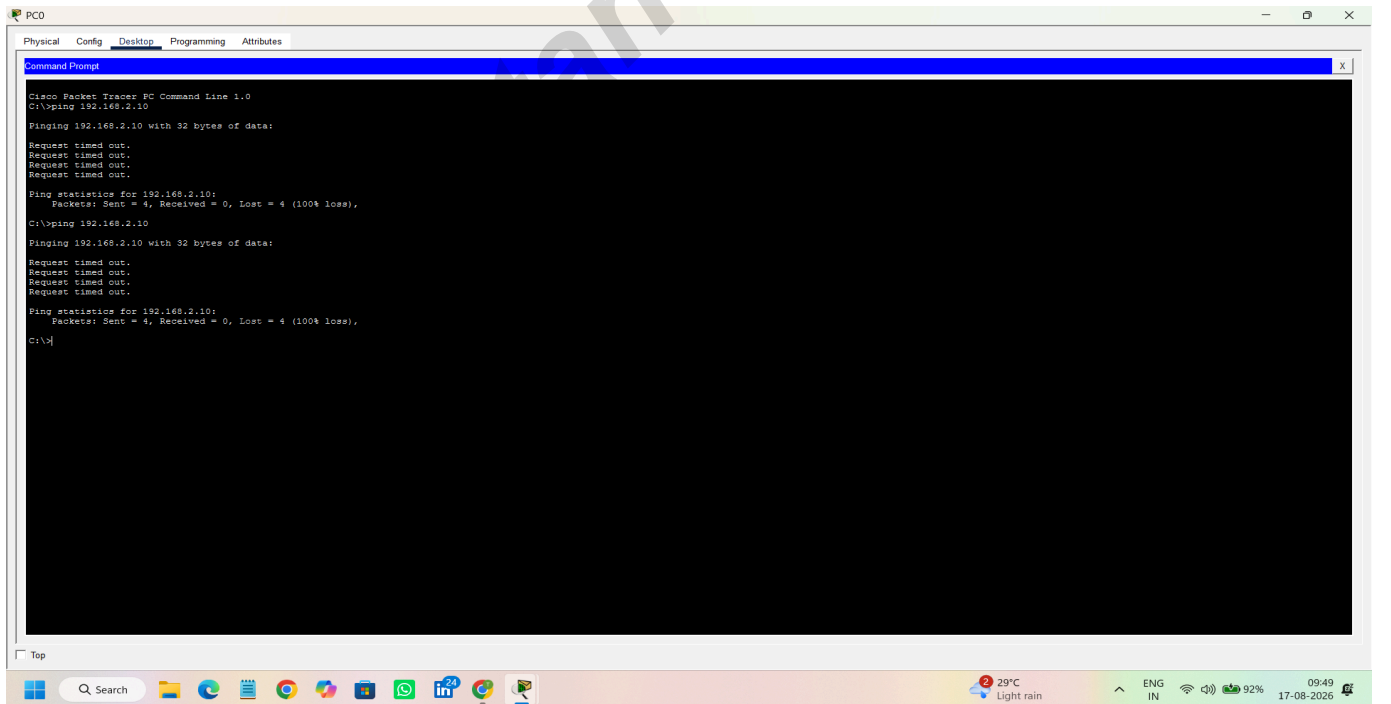
Router(config-iseekmp)#group 2
Router(config-iseekmp)#lifetime 86400
Router(config-iseekmp)#exit
Router(config)#crypto isakmp key cisco123 address 10.0.0.1
Router(config)#

Router con0 is now available.

Press RETURN to get started.

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 100 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
Router(config)#crypto ipsec transform-set VPN-SET esp-ess esp-sha-hmac
Router(config)#crypto map VPN-MAP 10 ipsec-iseekmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Router(config-crypto-map)#set peer 10.0.0.1
Router(config-crypto-map)#set transform-set VPN-SET
Router(config-crypto-map)#match address 100
Router(config-crypto-map)#exit
Router(config)#interface gigabitEthernet 0/1
%Invalid interface type and number
Router(config)#crypto map VPN-MAP
% Incomplete command.
Router(config)#exit
Router#
%SYS-5-CONFIG_I_1: Configured from console by console
```

Step 15:- Generate traffic



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

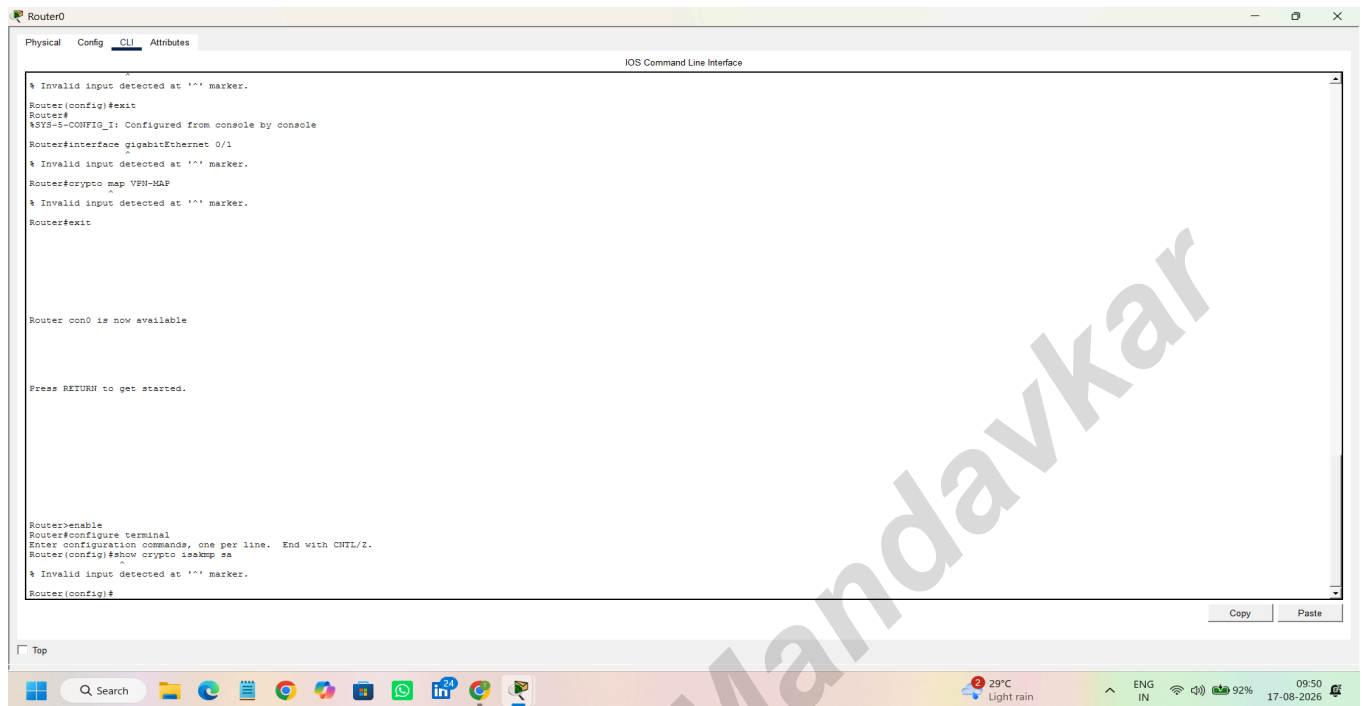
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Step 16:- Verify IKE



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

% Invalid input detected at '^' marker.
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#interface gigabitEthernet 0/1
% Invalid input detected at '^' marker.
Router#crypto map VPN-MAP
% Invalid input detected at '^' marker.
Router#exit
Router#exit

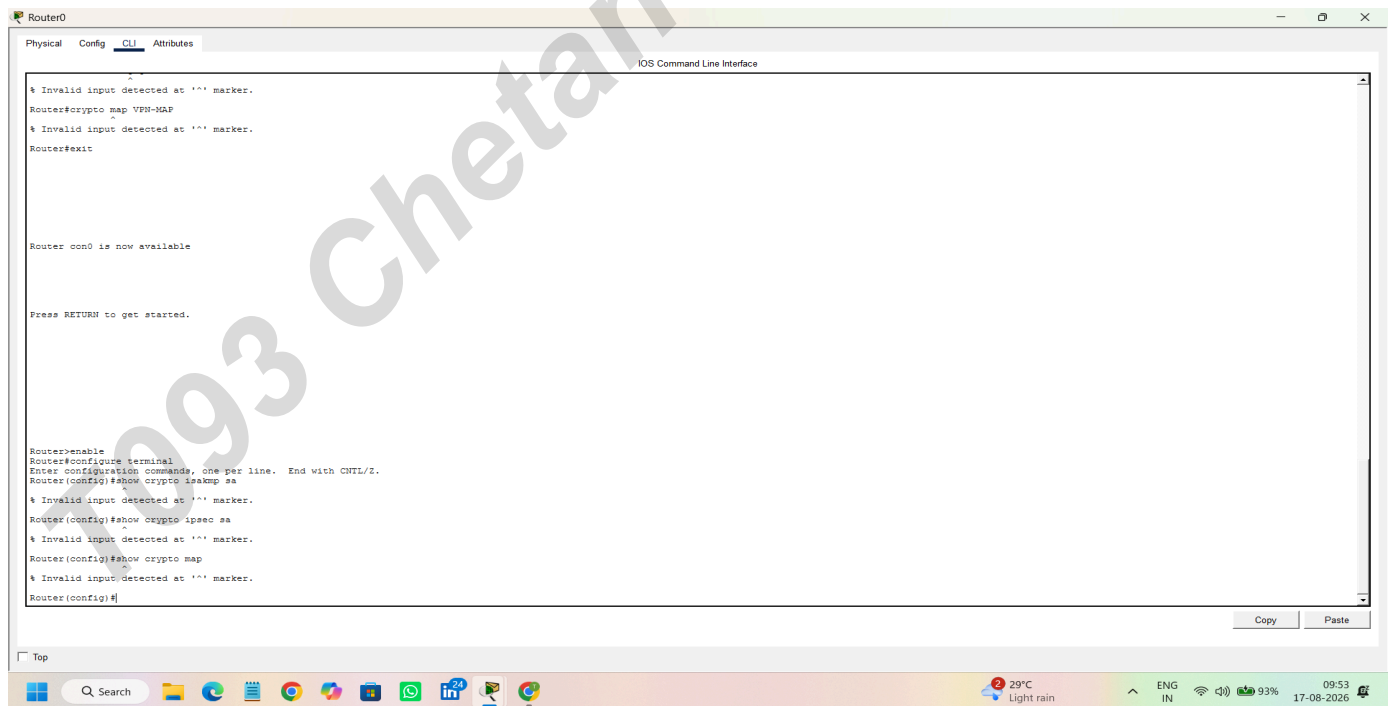
Router con0 is now available

Press RETURN to get started.

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#show crypto isakmp sa
% Invalid input detected at '^' marker.
Router(config)#

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#show crypto ipsec sa
% Invalid input detected at '^' marker.
Router(config)#show crypto map
% Invalid input detected at '^' marker.
Router(config)#
```

Step 17:- Verify IPsec



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

% Invalid input detected at '^' marker.
Router#crypto map VPN-MAP
% Invalid input detected at '^' marker.
Router#exit
Router#exit

Router con0 is now available

Press RETURN to get started.

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#show crypto ipsec sa
% Invalid input detected at '^' marker.
Router(config)#show crypto ipsec sa
% Invalid input detected at '^' marker.
Router(config)#show crypto map
% Invalid input detected at '^' marker.
Router(config)#
```

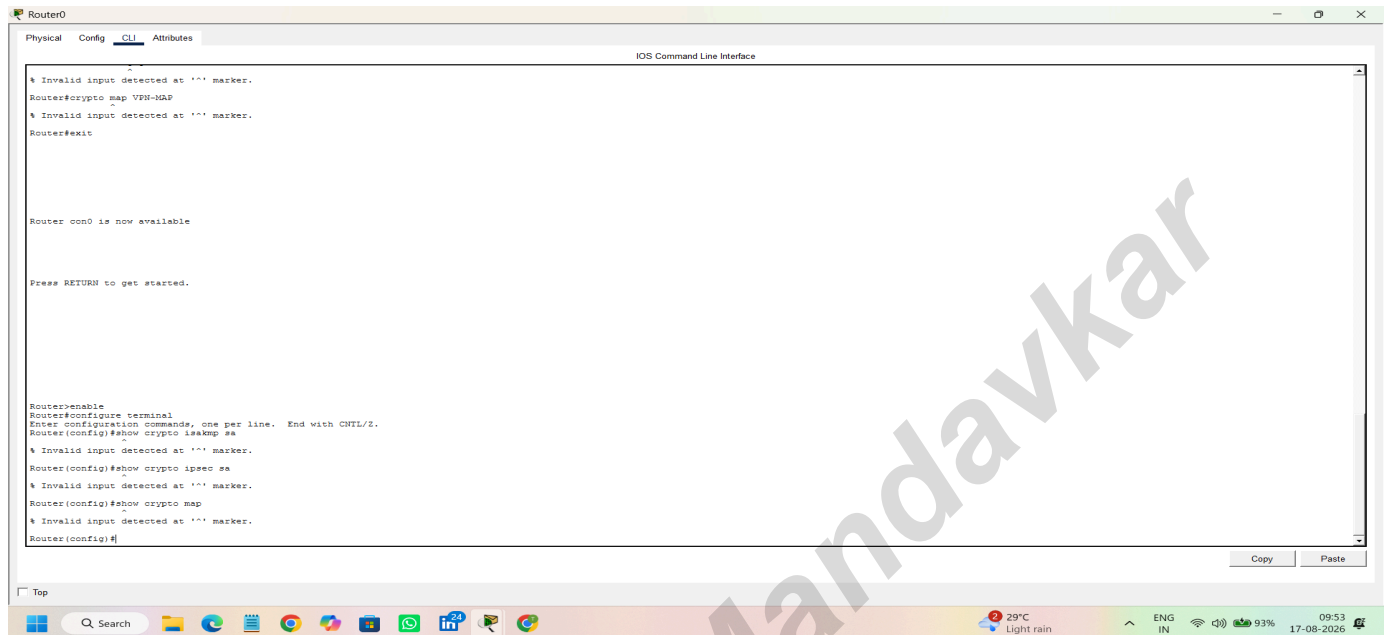
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Step 18:- Verify the Crypto Map



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router>
* Invalid input detected at '^' marker.
Router#crypto map VPM-SAP
* Invalid input detected at '^' marker.
Router#exit

Router con0 is now available

Press RETURN to get started.

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#show crypto isakmp sa
* Invalid input detected at '^' marker.
Router(config)#show crypto ipsec sa
* Invalid input detected at '^' marker.
Router(config)#show crypto map
* Invalid input detected at '^' marker.
Router(config)#
```

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